Course Syllabus

Course: 49-800

Title: Start Up Creation in Practice

Department: Integrated Innovation Institute

Semester: Fall 2026

Units: 6-12

Course Syllabus

Start Up Creation in Practice

Instructor: Sheryl Root (Silicon Valley)

Office: CMU SV Campus, B23, Rm 118

Email: sheryl.root@sv.cmu.edu

TA: TBD

Email: TBD

Office hours: By appointment/email

Class Meeting Logistics: *Fall 2026*

Note: Attending class meetings in person is required. Classes are recorded. Independent meetings for teams will not be recorded. Remote access will be provided pending professor approval.

Location: CMU SV Campus / Building 23 / Room 118

Day/Time: Tuesday/Thursday, 1530 to 1650 PT (3:30 PM to 4:50 PM Pacific Time)

Zoom:

============ Connection ===========
Zoom link TBD

Overview

The Startup Creation in Practice course takes a student's idea to commercialization. It gives students the opportunity to put into practice their new skills and knowledge gained at Carnegie Mellon to create a potential startup or have a startup they are trying to pursue. Students (and potential teams) work with the internal and external aspects of an extended project for a new business creation, large or small and the application of effective product management. They manage the viability, strategy, business model, risks, customer awareness, product management, quality and time dimensions of a project, as they interact with potential customers, stakeholders, business leaders, and technical personnel to achieve their stated objectives.

Course Description

Students will be solving a customer problem they have identified and as well as key elements to be defined including business requirements, user needs, technical obstacles, and create effective materials to fund their startups. This includes understanding the user, the market and the team necessary to deliver the solution that will succeed.

The course consists of reviewing and applying the concepts learned in the courses from the MSSM program including the courses in the Tech Venture concentration. These concepts are applied to the startup project for each team. Individuals will come prepared with their idea or startup concept for the course. The first day of each week will be focused to the discussion of content learned and what needs to be applied. The second day of each week will be focused on individual team discussion of application of concepts and progress with their projects. Students will be expected to create an initial requirements or MVP and measures to consider when fully implementing the solution to the problem they pose. Pitches for their final product will be presented in the last week of class over the two days of the course.

Learning Objectives

Students will be required to prepare and present their current idea, progress, and strategy for creating a product (MVP) and their potential business model and plan for execution at the end of the first 7 weeks. Students will be prepared to present their projects at the "Evening with Entrepreneurs" on October 29th.

At the end of Mini 2, the students will present their initial solutions (and/or MVP) to a panel of faculty to review the value and viability of their solutions.

For those graduating in December, 2026, you will have the potential to submit your proposal to the VB council in the Spring,. 2027.   The final presentation in Mini 2, 2026, should clearly demonstrate the following: business case (how you make money), initial market analysis (PMF and GTM), key metrics, personas, solution roadmap, and MVP if appropriate. This material will be the basis for your VB presentation. Students will have the opportunity to present their proposed startup at the Evening of Entrepreneurship in Mini 1, 2026. This will enable the introduction to investors, business leaders, and potential mentors/support.

Students will identify the team that created the opportunity and identify additional resources to go forward.

Learning objectives and deliverables for M1, M2 include the following:

1. Identify the potential business model (and alternatives considered) that will be most effective.
2. Create wire frames, MVPs, and/or basic prototypes in order to demonstrate potential solutions and test assumptions
3. Speak with a number of potential customers to determine user needs, process changes, and project constraints. Students will establish ability to create contextual design illustrations to validate customer and market.
4. Establish skills to interview selected key resources to assess validity and issues for the context of the potential solution.
5. If applicable, speak with developers in regards to technology and technical constraints.
6. Identify competitive environment and why the proposed idea/solution will work
7. Determine key risks, assumptions made or uncertainties to be resolved
8. Develop a method to track the success of the project with appropriate metrics and recommended actions.
9. Utilize a journal for tracking of activities, what was applied, what worked/did not work.

Readings: textbooks

1. (Strongly recommended) Yates, Linda K. *The Unicorn Within: How Companies Can Create Game-Changing Ventures at Startup Speed* ISBN-13: 978-1633698680

2. (Strongly recommended) Horine, Gregory M. *Project Management Absolute Beginner's Guide*. 4th Edition. ISBN-13: 978-0789756756
3. (Strongly recommended) Alexander Osterwalder et al. *Value Proposition Design: How to Create Products and Services Customers Want* ISBN-13: 978-1118968055
4. (Optional) Cagan, Marty *Inspired: How to Create Products Customers Love.* ISBN-13: 978-0981690407
5. BMG and Strategic Market Management texts from Spring BMS & AM courses .
6. Refresh the "The Lean Startup" readings and text from BMS course)
7. (Optional) Horowitz, Ben. *The Hard Thing about Hard Things: Building a Business When There are No Easy Answers.* March, 2014. ISBN-13: 978-0547265452
8. (Optional) Maurya, Ash. *Scaling Lean: Mastering the Key Metrics for Startup Growth.* June, 2016. ISBN-13: 978-1101980521
9. (Strongly recommended) The Launch Path by Bret Waters
10. (Strongly recommended) Aha by Andy Cunningham

Readings: articles

Individual readings will be provided from various sources such as McKinsey, Knowledge@Wharton, Stanford Business Journal, etc.

In addition, this course will use readings from *Harvard Business Review*. The CMU Library VPN provides full access to all HBR back issues. You need to log in with your Andrew ID.

1. Start at http://library.cmu.edu/
2. Under the 'Search & Find' tab, select 'Research databases'
3. In the 'All Subjects' dropdown box, select 'business'
4. Then click on the 'H' to see Harvard Business Review

Grading Components

Here are the general performance evaluation elements and their assessment weights.

The final course grade will be calculated using the following categories:

Assignment	**Percentage of Final Grade**
Project Plan & Weekly Journal	20%

Mid - Product Opportunity presentation materials	50%
Team & Product & Project Management	20%
Final pitch, Reflection & Learning document	10%

Project Plan & Weekly Journal: 20%. The plan of work describes the deliverables, project approach, roles/responsibilities, and project schedule for the startup.

Product Opportunity Presentation Materials: 50%. This describes the business opportunity, the business model to execute on the opportunity, the market fit, MVP, risks, assumptions, and next steps. The presentation should reflect the use of a storyboard and VC considerations to illustrate the opportunity in clear terms. The presentation should be no more than 15 minutes with 15 minutes for Q&A.

Team & Product and Project Management: 20%. The student team will decide which management roles and artifacts are needed to properly manage the work to achieve the expected results. Deciding the right approach requires trade-offs, and should ultimately be decided in light of the team's skills and needs. The project management approach should be documented in the plan. Included in this aspect of grading is the assessment of how the team worked together, managed challenges, adapted and revised efforts. The Product Management aspect will identify how the team adjusted based on customer interactions, reviews with faculty, and industry/technology changes. This would include any pivoting considerations for the product, how analysis was done and decisions made.

Startup Pitch Creation, Reflection & Learning: 10%. The team should prepare a 15-minute presentation for faculty (plus 15 minutes for Q&A for a total of 40 minutes). The purpose of the presentation is to identify how content from various courses was applied, share insights and lessons learned during the project, and identify what worked and didn't work during the development of the business opportunity. These lessons can include your reflections on managing people (customer interactions, team, or development support), on processes, and/or technology. The audience for the reflection presentation is CMU-SV faculty and those in the Startup Creation course. If there are interesting results you can share, you are free to include them, but the main purpose of this presentation is to reflect on successful product and project management practices.

High-Level Course Schedule (suggested)

The table below illustrates **some** of the potential topics as the course proceeds. The content may be adjusted if more in-depth discussion is required on some topics being applied. Each team will negotiate its own schedule, scope, and deliverables, based on the needs of their client and the project.

WEEK	**ACTIVITIES**
Week 1	Initial Meeting with all parties to cover content and expectations (faculty, students). Discuss proposed plan for managing the project process. Identify the current state of projects and establish timeline and deliverables during 14 weeks. No team meetings this week. Team members will sit together in classes. (Guest speaker: tbd)
Week 2	Customer development and business model materials completed. Preparation for potential mentorship event with Venture Bridge. Team meetings on second day of each week. (Guest speaker: open)
Week 3	Outline for final document and considerations for presentation. Further market research and analysis based on creation of MVP/prototype. Identifying market fit. (Guest speaker: tbd) Team Meetings on Thursday. **……..**
Week 7	Mid-Project Review: Presentation material to be completed: Solution concept, viability of problem being solved, customer identification, assumptions, uncertainties, risks, and Business Plan. In addition, if continuing in mini 2, complete Out-line for next 7 weeks. Presentations. For weeks 8 – 10 begin to develop product solution/MVP/wireframes.
Week 11	Communications, Basics in VC Pitch, Presentation Materials outline and expected content
Week 15	Presentations to Faculty and Guest Panel (End of Mini 2, Dec 12)*

(*) Last Week = last week of classes depending on semester breaks (may include finals week).

Students will be assigned the following final letter grades from the iii grading scale. All calculations for this grade will be based on details in the course assessment/deliverables section.

Grade	Percentage Interval
A	93-100
A-	90-92.9
B+	89-87.9
B	83-86.9
B-	80-82.9
C+	77-79.9
C	73-76.9
C-	70-72.9
D+	67-69.9
D	63-66.9
R	<=62

Grading Policies

- **Late-work policy**: There will be limited tolerance for late work. If late, submission must be within 24 hrs.
- **Make-up work policy**: Due to critical issues as recognized by faculty and TA, make up work will be identified and must be in writing, at least within 36 hours of the original due date,
- **Re-grade policy**: Requests must be in writing with detailed exceptions noted. Approved only by faculty.

iii Classroom Expectations

Punctuality

With the exception of part-time students at the Silicon Valley campus, students arriving more than 10 minutes late are marked as absent for the day. Students leaving more than 10 minutes early from class for meetings or other projects will also be marked absent for the day. Please discuss with your instructor(s) if you will have an unavoidable tardy or need to leave early.

iii Class Absences

Any student missing the first-class session of a required course will automatically drop one letter grade in addition to using up one of their "free pass" options to miss class. Except for the first day of class for a required course, students can miss class (a) 2 times per mini for sections that meet twice per week, (b) 1 time per mini for classes that meet once per week, and (c) 3 times per semester course that meet once or twice per week. These course absences will be considered as free pass options and will not reduce that student's grade. Additional absences lower their course grade by an amount determined by each faculty member for their own class. For their individual courses, instructors may specify dates for which attendance is necessary, and "free pass" options will not apply for these important dates to attend class.

Students are expected to communicate any absences (and all late arrivals) with their faculty members directly in advance of any events, when possible. It is recommended that students give at least 72-hours notice for a planned event. Notification of the absence does not mean it is approved. A faculty member will have final discretion on how late arrivals and additional absences might impact the course assignments, class participation and if additional options such as an Incomplete Grade should be considered.

Attendance

Students in full-time programs will be expected to attend courses that adhere to their designated modality, which for 2023/24 is in person expectation. Attendance policy changes and flexibility will be shared with students on an as needed basis.

Extracurricular Engagement

If students are interested in extracurricular engagement, the iii recommends a student limit their extracurricular activities to one per semester. These could include traveling to a conference, participating in a hackathon or design challenge or networking opportunities. The iii also recommends students look for leadership opportunities on campus in the forms of course project leads and GSA/Student Club leadership opportunity that would have only minimal impact on class attendance.

Except for the free pass options noted above, the iii does not approve absences for travel on non-program related events, career fairs or job interviews or student designed treks during the semester. As mentioned before, course instructors will have the final say regarding how student absences would be approved in their class.

Consent to Publicly Streaming Content

In order to protect the privacy of our students and to comply with FERPA, anyone who wishes to publicly livestream a presentation (i.e. critiques, capstone presentations, reviews, qualifying oral exams, as well as thesis and dissertation defenses conducted virtually) for class requirements, and/or record/share that presentation must first complete the [Consent to Publicly Livestream a Presentation form](#).

Academic Integrity, Collaboration & Cheating

Students are required to be fully aware of Carnegie Mellon University's policies regarding **Academic Integrity:**

https://www.cmu.edu/student-affairs/ocsi/students/avoiding/index.html

https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html

We would like to promote a collaborative environment where people feel free to openly discuss and ask questions. However, when assignments and projects are submitted, the work must be the individual's own. Simply put, cheating is submitting work that is not your own; material handed in for grading must be the product of the team for team deliverables and the product of the individual for individual deliverables; anything else constitutes cheating. Cheating in any form or shape will result in a failing grade for the course.

Generative AI Policy

To be determined

Accommodations for students with disabilities:

If you have a disability and have an accommodations letter from the Disability Resources office, I encourage you to discuss your accommodations and needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu.

Statement on Diversity and Inclusion:

We must treat every individual with respect. We are diverse in many ways and this diversity is fundamental to building and maintaining an equitable and inclusive campus community. Diversity can refer to multiple ways that we identify ourselves, including but not limited to race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, creed, ancestry, belief, veteran status, or genetic information. Each of these diverse identities, along with many others not mentioned here, shape the perspectives our students, faculty, and staff bring to our campus. We, at CMU, will work to promote diversity, equity and inclusion not only because diversity fuels excellence and innovation, but because we want to pursue justice. We acknowledge our imperfections while we also fully commit to the work, inside and outside of our classrooms, of building and sustaining a campus community that increasingly embraces these core values.

Each of us is responsible for creating a safer, more inclusive environment. Unfortunately, incidents of bias or discrimination do occur, whether intentional or unintentional. They contribute to creating an unwelcoming environment for individuals and groups at the university. Therefore, the university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and support, within the moment of the incident or after the incident has passed. Anyone can share these experiences using the following resources:

Center for Student Diversity and Inclusion

csdi@andrew.cmu.edu, (412) 268- 2150

Report-It online anonymous reporting platform:

reportit.net username**:** *tartans* password**:** *plaid*

All reports will be documented and deliberated to determine if there should be any following actions. Regardless of incident type, the university will use all shared experiences to transform our campus climate to be more equitable and just.

Student Well-Being

Make sure to move regularly, eat well, and reach out to your support system or me **[sheryl.root@west.cmu.edu]** if you need to. We can all benefit from support in times of stress, and this semester is no exception.

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. CMU services are available, and treatment does work.

Pittsburgh Campus

Counseling and Psychological Services (CaPS): 412-268-2922

Learn more about confidential mental health services available on campus at

http://www.cmu.edu/counseling/. Support is always available (24/7).

University Health Services: https://www.cmu.edu/health-services/about-us/index.html

Silicon Valley Campus

Statement on Student Wellness & Resources: https://docs.google.com/document/d/1HfPonlzhfoVABHycJxynYX7nE9Qj9hyoz2U2-WJ-5-A/edit

Links to an external site.

Use of technology during class

For some class sessions, we will be using **zoom to enable teaming and remote guest speakers/students** which can be accessed on a smartphone or laptop. If you do not have the necessary equipment, please contact your HUB liaison who is available to help you tap into appropriate resources.